

Capstone Project

Project Title:

Automation Testing of E-Commerce Web Application using Playwright

Objective

- Automate end-to-end testing using Playwright
- Achieve minimum 120 test cases
- Cover at least 8 modules/services
- Implement Page Object Model (POM)
- Generate Allure & HTML reports
- Capture screenshots and videos for failures

Technologies and Tools

Category	Tools / Technology
Automation Tool	Playwright
Language	JavaScript
Framework	Playwright Test Runner
Design Pattern	Page Object Model (POM)
Reporting	Allure Report, HTML Report
IDE	VS Code

Modules / Services

1. **User Authentication Service** : verifies user identity through sign up, login, and logout functionalities.
2. **Product Catalog Service** : whether all product categories and their respective products are displayed correctly with accurate details like name, image, and price. We will also verify category switching, filtering, and proper loading of product listings without errors.
3. **Product Details Service** : We will test the product details page to ensure correct display of product information such as name, price, image, and description.
4. **Shopping Cart Service** : We will test the shopping cart functionality to ensure products can be added, viewed, updated, and removed correctly.

5. **Checkout Service :** We will test the checkout process to ensure users can place orders by entering valid details in the order form. We will also verify form validations, payment information handling, and successful order confirmation.
6. **Order Management Service :** We will test order creation and confirmation to ensure a successful order generates an order ID and confirmation message
7. **Navigation Service :** We will test all navigation links to ensure users can move between Home, Cart, Contact, About, Login, and Signup pages correctly.
8. **Contact / Support Service :** We will test the contact form to ensure users can successfully send messages with valid input. We will also verify form validations, alert handling, and proper submission behavior for both valid and invalid inputs

Project Goals and Deliverables

- Develop at least 120 automated test scenarios covering major application flows
- Design and implement a minimum of 8 functional modules/services within the test framework
- Integrate Allure and HTML reporting for detailed test execution results
- Ensure cross-browser testing support (Chromium, Firefox, WebKit)
- Implement parallel test execution to reduce overall runtime
- Follow clean code practices and modular folder structure for better readability

Severity and Test Cases :

Service / Module	Severity	Test Cases
User Authentication	Critical	15
Product Catalog	High	15
Product Details	High	15
Shopping Cart	Critical	20
Checkout	Critical	15
Order Management	High	10

Navigation Service	Medium	15
Contact / Support	Low	15

Conclusion:

This project builds a scalable Playwright automation framework to test an e-commerce application with strong coverage of key user workflows using the Page Object Model.